

PHYSICS — SSC-I (Class 9) — Marking Report

Exam-02 · Chapters 3, 4, 9 (Dynamics-I, Dynamics-II, Nature of Science) · Version 2 · Attempted 10 April 2026

Student: Seemab · Date Marked: 10 April 2026 · Total Marks: 65 · Time: 2 h 30 min

12 / 12 SECTION A · MCQS	30.5 / 33 SECTION B · SHORT	19.5 / 20 SECTION C · LONG	62 / 65 TOTAL · 95.4% · A+
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SECTION A — Multiple Choice Questions (12 / 12)

#	Question (short)	Correct	Seemab	Mark
1	Non-contact force	C — Gravitational force	C	1
2	Newton's first law also called law of	B — Inertia	B	1
3	SI unit of force ($F = ma$)	B — $\text{kg}\cdot\text{m}/\text{s}^2$	B	1
4	Action and reaction forces act on	B — Different bodies	B	1
5	SI unit of momentum ($p = mv$)	C — $\text{kg}\cdot\text{m}/\text{s}$	C	1
6	Weakest fundamental force	D — Gravitational	D	1
7	Torque = force $\times$?	C — Moment arm	C	1
8	For complete equilibrium, both must be zero	B — Net force & net torque	B	1
9	Cone resting on its base is in	A — Stable equilibrium	A	1
10	Which does NOT reduce friction	D — Increasing roughness	D	1
11	If speed doubled in $F_c = mv^2/r$	C — Four times	C	1
12	Tentative assumption verifiable is a	C — Hypothesis	C	1

Section A: 12 / 12 (100%) — Perfect MCQ score. Strong conceptual mastery across Dynamics-I, Dynamics-II, and Nature of Science.

SECTION B — Short Answer Questions (30.5 / 33)

Q	Topic	Marks	Comment
2(i)	Force; contact vs non-contact forces	3 / 3	Clear definition; cricket bat & Earth-Moon gravity examples are excellent.
2(ii)	Newton's 1st law & inertia	3 / 3	Statement, soccer ball example, and inertia definition all complete.
2(iii)	Newton's 2nd law; bus $a = F/m$	3 / 3	$a = 3000 / 12000 = 0.25 \text{ m/s}^2$. Minor: wrote "0.25 m/s" — unit slip (m/s^2 intended).
2(iv)	Momentum; golf ball $p = mv$	3 / 3	$0.045 \text{ kg} \times 80 \text{ m/s} = 3.6 \text{ kg}\cdot\text{m/s}$. Conversion shown clearly.
2(v)	Conservation of momentum + math form	2.5 / 3	Statement and condition fine. Math form was given as $P_f = P_i / m_1v_1 = m_2v_2$. For a two-body system the form should be $m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$. -0.5
2(vi) OR	Principle of moments — seesaw	3 / 3	$250 \times 0.6 = 200 \times d \rightarrow d = 0.75 \text{ m}$. Final unit slip "0.75 N·m" — should be "m" (metres). Accepted.
2(vii)	First & second conditions of equilibrium	2.5 / 3	Both conditions stated with formulas. Did not define <i>equilibrium</i> itself, nor explicitly note that both conditions must hold simultaneously for complete equilibrium. -0.5
2(viii) OR	Three methods of reducing friction	3 / 3	Polishing, ball bearings, lubricants — all explained correctly.
2(ix)	Terminal velocity & skydiver	1.5 / 3	Defined terminal velocity correctly. Skydiver explanation missed the key physics: drag force grows until it equals weight, so net force = 0 and $a = 0 \rightarrow$ constant velocity. Instead wrote about skydivers manipulating posture/drag — that explains <i>changing</i> speed, not <i>why</i> they reach terminal velocity. -1.5
2(x) OR	Orbital speed; why same v for all sats	3 / 3	Full derivation: $mv^2/r = GMm/r^2 \rightarrow v^2 = GM/r$, satellite mass cancels. Excellent.

2(xi)	Physics & four branches	3 / 3	Definition + Classical, Climate, Particle, Thermal physics — all defined.
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Section B: 30.5 / 33 (92.4%) — Strong across the board. Two small process slips and one incomplete physics explanation.

SECTION C — Detailed Answer Questions (19.5 / 20)

Q	Topic	Marks	Comment
3	Newton's 3 laws + 2 limitations	5 / 5	Each law stated, mathematical form given, example provided. Limitations: atomic scale (quantum mechanics) and high speeds (relativity). Complete.
4	Momentum, impulse, derivation, F to stop car & truck	4.5 / 5	Definitions and impulse-momentum derivation correct. Arithmetic error in (b)(i): car force computed as -1020 N instead of the correct -1920 N ($9600 / 5 = 1920$, not 1020). Truck calculation ($-16,000 \text{ N}$) is correct. -0.5
5	Equilibrium · principle of moments verification · types & CG	5 / 5	Both conditions of equilibrium with formulas. Verification: $4 \times 0.2 = 2 \times 0.4 \rightarrow 0.8 \text{ N}\cdot\text{m} = 0.8 \text{ N}\cdot\text{m}$. Stable / unstable / neutral with role of CG — all complete.
6	Centripetal force · hammer throw · Earth orbital speed	5 / 5	$F_c = 5 \times 25^2 / 1.5 = 2083.3 \text{ N} \checkmark$. Earth orbital speed: $T = 3.156 \times 10^7 \text{ s}$, $r = 1.5 \times 10^{11} \text{ m}$, $v = 2.98 \times 10^4 \text{ m/s} \approx 29.8 \text{ km/s} \approx 107,280 \text{ km/h} \checkmark$. Excellent.

Section C: 19.5 / 20 (97.5%) — Outstanding long-answer performance. Only one arithmetic slip in the car braking force.

Final Result

62 / 65 TOTAL SCORE	95.4 % PERCENTAGE	A+ GRADE	3 MARKS LOST
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Strengths

- **Perfect MCQ score** — strong grasp of dynamics, equilibrium, and Nature of Science concepts.
- **Clean numerical working** — momentum, hammer throw, Earth orbital speed all flawless with units shown.
- **Strong derivations** — impulse-momentum relation and $v^2 = GM/r$ both fully derived from first principles.
- **Newton's laws of motion** — all three laws stated precisely with math forms and clear examples.
- **Principle of moments** — verification done correctly with $\text{N}\cdot\text{m}$ balanced on both sides.

Areas to Improve

- **Arithmetic checking (Q.4 b-i):** $9600 / 5 = 1920$, not 1020. Always recheck the final division step. **-0.5**
- **Terminal velocity physics (Q.2 ix):** The key idea is *drag force grows with speed until it equals the weight*; at that point $F_{\text{net}} = 0$ and $a = 0$ — so velocity stays constant. Memorise this drag-equals-weight balance. **-1.5**
- **Conservation of momentum form (Q.2 v):** for a two-body collision, write $m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$, not just $P_f = P_i$. **-0.5**
- **Equilibrium definition (Q.2 vii):** always start with "A body is in equilibrium when net force AND net torque are both zero" before listing the two conditions. **-0.5**
- **Unit discipline:** two small slips — "0.25 m/s" instead of m/s^2 (Q.2 iii), and "0.75 $\text{N}\cdot\text{m}$ " instead of m (Q.2 vi). Not deducted, but examiners can penalise.

Compared to Previous Physics Exam

Exam	Date	Section A	Section B	Section C	Total	%
exam-01-ch1-2 (Measurement, Kinematics)	7 Apr 2026	12 / 12	32.5 / 33	19 / 20	63.5 / 65	97.7 %
exam-02-ch3-4-9 (Dynamics, Nature of Science)	10 Apr 2026	12 / 12	30.5 / 33	19.5 / 20	62 / 65	95.4 %

Slight dip from 97.7 % $\rightarrow$ 95.4 %, all attributable to a single arithmetic error and one incomplete conceptual answer (terminal velocity). MCQ and long-question performance remain perfect / near-perfect.